A Practitioner's Guide to Inference with Spatial Dependence

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Abstract. In many empirical applications in the social sciences, individuals-level observations interact or are subject to shared shocks across space. In order to conduct valid statistical inference, such spatial dependence must be accounted for. While a number of strategies to account for spatial dependence exist, these typically are based on strict assumptions about delimitation of dependence within certain geographic areas (*i.e.* clustering), or modelling dependence solely in terms of distance. In this paper, we discuss these common forms of inference with spatial dependence and their drawbacks, and introduce an alternative form of inference, laid out in DellaVigna et al. (2025), known as Thresholding Multiple Outcomes (TMO). This procedure for inference models dependence by using information from a range of variables to infer the dependence between units across space, and constructs standard errors and confidence intervals based on such dependence. We introduce the command `tmo`, which implements these procedures in Stata for a range of linear models including OLS and 2SLS.

Keywords: spatial dependence, inference, multiple outcomes, standard errors.

1 Introduction

Spatial dependence is a common feature of data in the social sciences. Interactions between individuals in their neighbourhoods, cities, states or other social networks typically imply that individuals cannot be viewed as independent observations in space. And even absent direct interactions between individuals, shared underlying shocks typically occur in areas which are linked across potentially quite broad areas of space, and potentially in quite a complex fashion. Productivity shocks in one region propagate to others through trade and labour-market linkages, policies adopted in one jurisdiction shape both policy adoption and outcomes elsewhere, and common shocks—weather events, commodity price movements, health crises—strike collections of locations that need not be contiguous. Importantly, the locations which co-move need not be near neighbours: two distant, but economically similar, agricultural counties may respond to a commodity price shock far more similarly than two adjacent counties, one urban and one rural.

Such spatial dependence must be accounted for if one wishes to conduct inference on parameters estimated in econometric models. Variance-covariance estimators in commonly estimated econometric models such as ordinary least squares (OLS) and two stage least squares (2SLS) typically start from the assumption of independence, but must be relaxed to account for more realistic interactions among observations across space. If such adjustments are not made, standard errors will be mis-estimated, generally resulting in standard errors which are too small, and confidence intervals which are too narrow.

In this paper we lay out considerations for conducting inference in models where spatial dependence among observations is suspected. We discuss a number of existing common procedures and their limitations, and lay out a new procedure for inference with spatial correlation variance which leverages multiple outcome measures. This procedure, referred to as Thresholding Multiple Outcomes (TMO) is described in DellaVigna et al. (2025) (and in this paper), and is accompanied by a Stata routine `tmo` designed for a range of commonly encountered estimation methods in Stata which we document here.

Tools to relax the variance-covariance matrix *in certain ways* are default in many computational implementations in Stata. For example, essentially all Stata estimators incorporate heteroscedasticity-robust variance estimators. Similarly, cluster-robust variance estimation allows for *limited* spatial dependence: observations may correlate arbitrarily within a designated group—a state, say—but are assumed independent across groups. The consequences of ignoring such intra-group correlation have long been understood (Moulton 1986, 1990), and clustering is now routine empirical practice.

However, clustering assumes a very specific form of dependence, in which correlation stops abruptly at the boundary of the cluster. A second family of procedures instead models dependence explicitly as a function of distance, either geographic or 'economic'. Conley (1999) proposes a kernel-based estimator in which the covariance between two locations is down-weighted as the distance between them grows, reaching zero at a chosen cutoff. More recently, Müller and Watson (2022b,a) propose Spatial Correlation

Principal Components (SCPC), which conducts inference using a small number of low-frequency principal components derived from a worst-case spatial correlation model, offering improved size control when observations agglomerate in space.

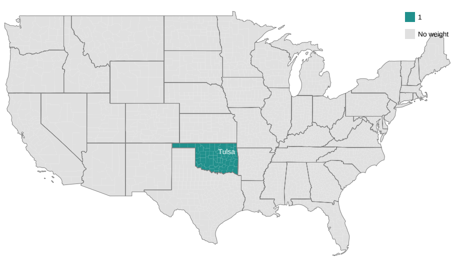
Both families, however, impose *a priori* that dependence is organised by geography: either delimited by borders, or declining in distance. It is reasonable to ask what dependence looks like in practice. DellaVigna et al. (2025) propose to answer this question empirically: when a researcher observes many outcomes for the same units—as is very often the case—those auxiliary outcomes can be used to *estimate* which pairs of locations are correlated, rather than assuming where correlation lies. Their procedure, Thresholding Multiple Outcomes (TMO), then conducts inference on the parameter of interest allowing for dependence among precisely those pairs.

To see how these approaches differ, consider a representative county: Tulsa, Oklahoma. Figure 1 displays, for each of the four procedures discussed above, which other counties enter the variance calculation for a regression using county-level data, and with what weight. Clustering by state (panel a) admits arbitrary correlation between Tulsa and the rest of Oklahoma, and nothing else. The Conley approach (panel b) weights counties by a declining function of their distance to Tulsa, reaching zero at 300 miles: dependence is assumed strongest nearby, and absent far away. SCPC (panel c) is displayed through its implicit pair-level kernel: because inference is based on a handful of smooth, low-frequency spatial components, *every* county in the sample receives some weight, largest in a broad region around Tulsa and varying smoothly over the whole map—there is no threshold and no cutoff. TMO (panel d) is the mirror image: using 90 auxiliary county-level outcomes, it identifies the small set of counties whose outcomes systematically co-move with Tulsa's (about 0.5% of all pairs sample-wide), and allows arbitrary correlation with those counties only—wherever they happen to be on the map, including far beyond any distance cutoff.

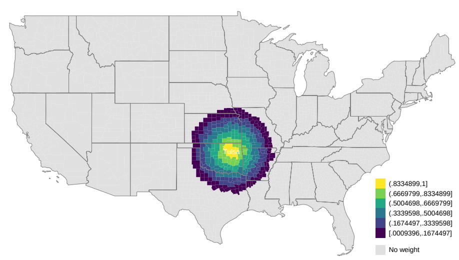
In this paper we provide a primer on estimating standard errors under spatial dependence in Stata, introduce the TMO procedure of DellaVigna et al. (2025) and its implementation in the `tmo` command, and work through a series of examples based on a rich collection of U.S. county-level outcomes. A distinguishing feature of the command is that TMO need not displace existing practice: it can *augment* clustered, distance-based, or SCPC inference, adding to the variance calculation those pairs of locations whose auxiliary outcomes reveal dependence that the baseline procedure would miss. We document these combinations, alongside cross-sectional, instrumental variables, and panel implementations.

A number of related tools exist for spatial inference in Stata, including implementations of Conley standard errors, the `scpc` command of Müller and Watson, and the `spur` suite of Becker et al. (2026) for diagnosing spatial unit roots. The `tmo` command complements these by providing inference based on estimated, rather than assumed, patterns of cross-location dependence.

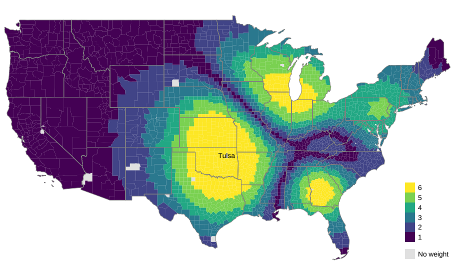
In what remains of this paper, we begin by describing methods in Section 2, first discussing proposals to deal with spatial dependence, before focusing on TMO more generally. We discuss a Stata command to implement TMO in Section 3, before docu-



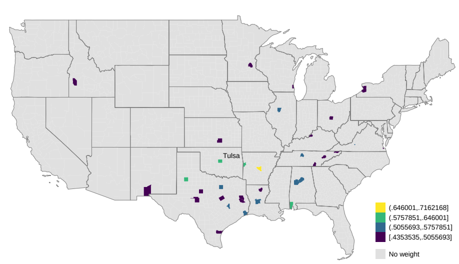
(a) Clustering by State



(b) Conley



(c) SCPC



(d) TMO

Figure 1: Spatial Standard Error Procedures

menting a number of examples of the use of this command—`tmo`—in Section 4. Section 5 lays out key points of relevance for practitioners and concludes.

2 Methods

2.1 A Simple Bivariate Regression Model

Consider estimating a simple bivariate regression model of the type:

$$Y_i = \alpha + \tau W_i + \varepsilon_i, \tag{1}$$

based on a sample of data $i = 1, \ldots, N$, where Y_i is some outcome of interest, W_i some treatment or independent variable of interest and ε_i an unobserved term. Assume now that $\mathbb{E}[\varepsilon_i|W_i] = 0$. In this setting, the parameters α and τ can be estimated consistently by ordinary least squares, and denoted $\widehat{\alpha}$ and $\widehat{\tau}$ respectively. Here—and in most empirical settings—we are most interested in the quantity $\widehat{\tau}$.

While the estimation of $\widehat{\tau}$ is straightforward, the estimation of the standard error for this estimate hinges upon assumptions about the correlation among residuals ε_i within the sample. To see this, note that the standard error is the square root of the following variance term:

$$\mathbb{V}(\widehat{\tau}|W) = (W'W)^{-1}W'\Omega W(W'W)^{-1}. \tag{2}$$

This depends, crucially, on the matrix Ω which is an $N \times N$ matrix describing the covariance between each ε_i in the sample. The most restrictive assumption about this matrix is that all observations share a common variance, and all observations are independent, or in other words, observations are independent and identically distributed. In this case, Ω simplifies to $\sigma^2 I$, with σ^2 referring to the variance of Y_i, and I and $N \times N$ identity matrix. This leads to the simple variance expression:

$$\mathbb{V}(\widehat{\tau}|W) = \sigma^2(W'W)^{-1}, \tag{3}$$

which is returned as default in computational estimators of (1).

The assumption of a common variance can be loosened quite simply to reflect heteroscedasticity, as laid out in seminal work of Huber (1967); Eicker (1967); White (1980). In this case, rather than (3), the heteroscedasticity-robust analogue:

$$\widehat{\mathbb{V}}(\widehat{\tau} \mid W)_H = (W'W)^{-1}\left(\sum_{i=1}^{N}\widehat{\varepsilon}_i^2 W_i W_i'\right)(W'W)^{-1} \tag{4}$$

can be estimated, where $\widehat{\varepsilon}_i = Y_i - \widehat{\alpha} - \widehat{\tau}W_i$. Again, such a variance estimator is standard in most computational implementations of regression estimators, such as Stata's `robust` option.[1] However, both (3) and (4) are predicated on the assumption that observations are independent, and as such ε_i and ε_j are not correlated for all observation pairs i and

[1]In practice, because of finite sample biases in this estimator, most software packages report finite-sample corrected versions of the heteroscedasticity-robust variance estimator. Stata's `robust` option

j where $i \neq j$. In many cases, this is likely viewed as an unrealistic assumption, and instead some sort of spatial (or other) correlation is presumed to exist between terms ε_i and ε_j. In this case, a number of procedures exist to seek to conduct appropriate inference despite such dependence.

Restricting Elements of Ω

A first way to proceed is to note that we can permit limited sorts of spatial dependence by clustering at some aggregate geographic level. This recognises the importance of intra-group clustering (Moulton 1986), by allowing certain dependence across space within Ω while restricting elements of Ω outside of any specific cluster to be 0. In essence, this allows arbitrary correlations within some specific geographical or other boundary, while restricting any dependence across clusters. Such a case can be viewed schematically in Ω_{CR} below, for a cluster-robust case assuming a number of blocks, compared to the more restrictive heteroscedasticity-robust option Ω_H.

$$\Omega_{\mathrm{CR}} = \begin{bmatrix} \varepsilon_1^2 & \varepsilon_1\varepsilon_2 & 0 & 0 & \cdots & 0 & 0 \\ \varepsilon_2\varepsilon_1 & \varepsilon_2^2 & 0 & 0 & \cdots & 0 & 0 \\ 0 & 0 & \varepsilon_3^2 & \varepsilon_3\varepsilon_4 & \cdots & 0 & 0 \\ 0 & 0 & \varepsilon_4\varepsilon_3 & \varepsilon_4^2 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & 0 & \cdots & \varepsilon_{N-1}^2 & \varepsilon_{N-1}\varepsilon_N \\ 0 & 0 & 0 & 0 & \cdots & \varepsilon_N\varepsilon_{N-1} & \varepsilon_N^2 \end{bmatrix} \Omega_{\mathrm{H}} = \begin{bmatrix} \sigma_1^2 & 0 & \cdots & 0 \\ 0 & \sigma_2^2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \sigma_N^2 \end{bmatrix}$$

Inference based on clustering is standard, following Liang and Zeger (1986); MacKinnon and White (1985),

$$\widehat{\mathbb{V}}(\widehat{\tau} \mid W)_{CR} = (W'W)^{-1} \left(\sum_{g=1}^{G} W_g' \widehat{\varepsilon}_g \widehat{\varepsilon}_g' W_g \right) (W'W)^{-1} \tag{5}$$

where the g refers to each of the G clusters, $\widehat{\varepsilon}_g$ a vector of residuals for each cluster, In practice, Stata's implementation of such cluster-robust variance estimates incorporates a finite sample correction factor, multiplying (5) by $\frac{G}{G-1}\frac{N-1}{N-K}$ to correct for the downward bias in standard errors when the number of clusters is small (Cameron and Miller 2015; Rogers 1993).

Distance-based limiting

A drawback of such an approach is that spatial dependence is viewed as being sharply demarcated by the borders or limits of clusters. Another common alternative which

reports the HC1 estimator by default, which rescales the HC0 matrix in (4) by $\frac{N}{N-k}$, where k is the number of estimated parameters (2 in this case). Other common variants include HC2 and HC3, which apply leverage-based corrections to adjust for influential observations (see e.g. MacKinnon and White 1985; Long and Ervin 2000), also available via Stata's vce() option.

seeks to avoid such sharp demarcation at geographic borders instead views spatial dependence in terms of distance. This approach, proposed by Conley (1999), allows for residual dependence to decay continuously over space. Rather than imposing a block-diagonal structure in Ω, Conley's estimator permits non-zero covariances between all observations but downweights them by a kernel function of geographic distance, up to some maximum point at which case zero covariance is assumed.

Formally, the Conley variance estimator takes the form:

$$\widehat{\mathbb{V}}(\widehat{\tau} \mid W)_{\text{Conley}} = (W'W)^{-1} \left(\sum_{i=1}^{N} \sum_{j=1}^{N} K_{ij} \widehat{\varepsilon}_i \widehat{\varepsilon}_j W_i W_j' \right) (W'W)^{-1}, \tag{6}$$

where $K_{ij} = K(d_{ij}/b)$ is a kernel function of the distance d_{ij} between units i and j, and b is a bandwidth parameter. Typical choices for $K(\cdot)$ include the Bartlett or uniform kernel.[2] Intuitively, this estimator treats spatial correlation as strongest between nearby units, either uniformly, or declining smoothly with distance, and ultimately reaching zero beyond a certain cutoff. We can see intuitively that as K_{ij} decays to zero, the correlation between residuals in (6) becomes irrelevant in the variance estimate, as the summation term resolves to 0 when $K_{ij} = 0$.

There are also alternative approaches which seek to take such-distance based approaches. Müller and Watson (2022b) point out that if observations are not uniform in space, then methods such as (6) will not be asymptotically valid, and potentially suffer from size distortions in finite samples which lead to over-rejection of null hypotheses. Thus, this leads to suggestions as laid out in Müller and Watson (2022b,a) in which rather than using kernel functions to model the location of units across space, the location of units is modelled based on principal components which capture spatial agglomeration. They show that this approach offers improved performance in cases where spatial agglomeration of observations occurs, and, when confidence intervals are constructed based on critical values which consider such spatial dependence, can result in large improvements in error rates (Müller and Watson 2022a).

Thresholding Multiple Outcomes

Using Multiple Outcomes to Infer Cross-Space Correlations Ultimately, however, both cluster-robust and distance-based procedures are limiting, either by assuming a strict off-diagonal zero correlation, or by assuming a zero correlation among units beyond some specified cut-off point. This limitation occurs given that these approaches are based entirely upon modelling spatial correlations by considering how *units* relate across space, rather than considering how outcomes Y_i themselves relate across space. Thus, a more realistic approach may be to seek to infer geographic dependence by examining the correlation of a range of actual measures over space.

[2]These are defined respectively as $K(d_{ij}/b) = 1 - d_{ij}/b$ for $d_{ij} \leq b$ and 0 otherwise, which applies linearly decreasing weights as distance increases; and $K(d_{ij}/b) = \mathbb{1}\{d_{ij} \leq b\}$, which applies equal weight to all pairs within a cutoff distance b.

To this end, DellaVigna et al. (2025) introduce Thresholding Multiple Outcomes, or TMO, in which auxiliary measures are used to determine whether correlations exist between particular locations in data. This requires information on a number d of auxiliary outcomes, which we will denote $Y_i^{(1)}, \ldots, Y_i^{(d)}$. If such multiple alternative outcomes exist, we can use these measures to assess whether systematic correlation exists between residuals over space.

Namely, each of these outcomes which we will generically refer to as $Y_i^{(j)}$ can themselves be regressed on our outcome of interest in (1). The residual term from this regression is denoted $\widehat{\varepsilon}_i^{(j)}$, and this can be normalised to have a variance of 1 as $\tilde{\varepsilon}_i^{(j)} = \widehat{\varepsilon}_i^{(j)} / \sqrt{\frac{1}{N}(\widehat{\varepsilon}_i^{(j)})^2}$. We will seek to leverage these residuals to determine the correlation of outcomes between locations across space. To do so, the covariance and the correlation between residuals for some locations i and i' are denoted:

$$\widehat{\lambda}_{i,i'} = \frac{1}{d}\sum_{j=1}^{d}(\tilde{\varepsilon}_i^{(j)} - \bar{\varepsilon_i})(\tilde{\varepsilon}_{i'}^{(j)} - \bar{\varepsilon_{i'}}) \qquad \text{and} \qquad \widehat{\rho}_{i,i'} = \frac{\widehat{\lambda}_{i,i'}}{\sqrt{\widehat{\lambda}_{i,i}\widehat{\lambda}_{i',i'}}}, \tag{7}$$

where $\bar{\varepsilon_i} = \frac{1}{d}\sum_{j=1}^{d}\tilde{\varepsilon}_i^{(j)}$. This quantity $\widehat{\rho}_{i,i'}$ is of key importance, as it summaries, across all outcomes d, the tendency of residuals from observation i and i' to be correlated.

The logic of TMO is that while most pairs of units are not correlated, some will be, and the correlation between auxiliary outcomes $\widehat{\rho}_{i,i'}$ can be used to determine those observations which are correlated across space. Specifically, setting some threshold value δ as the value above which observations are inferred to be meaningfully correlated, we can write $\widehat{\Omega}$ as:

$$\widehat{\Omega} = \begin{cases} \widehat{\varepsilon}_i^0 \widehat{\varepsilon}_i^0 & \text{if } i = i', \\ \widehat{\varepsilon}_i^0 \widehat{\varepsilon}_{i'}^0 & \text{if } \widehat{\rho}_{i,i'} \geq \delta, \\ 0 & \text{if } \widehat{\rho}_{i,i'} < \delta. \end{cases}$$

In essence, this threshold term δ is used to indicate which units have a systematic correlation between residuals across space, and as such account for this correlation in variance estimates. Note that previously, off diagonal terms were $\widehat{\varepsilon}_i^0 \widehat{\varepsilon}_{i'}^0$ were allowed to be non-zero if they fulfilled specific *geographic* criteria such as sharing a cluster, whereas here with the incorporation of δ, these terms are allowed to be non-zero if *auxiliary outcomes themselves* are observed to be correlated across space.

DellaVigna et al. (2025) show that under two assumptions, these procedures result in a consistent variance estimate. Specifically, these assumptions are referred to as 'proportionality' and 'sparsity'. The first of these implies that the covariance of residuals for auxiliary outcomes is informative of the correlation across residuals for the outcome of interest. And the second assumption implies that there is some limit on the number of units which are actually correlated with any other unit.

Determining Thresholds for Cross-Space Correlations Provided that such assumptions are reasonable, the remaining question is what threshold value for δ should be used as a threshold for determining correlations. In effect, most pairs of observations will not be correlated ("null pairs"), while some small proportion ("non-nulls") will be correlated. Selecting a threshold thus can be viewed as a trade-off between potentially excluding relevant non-nulls (which will understate the variance) and including irrelevant nulls (which will overstate the variance). DellaVigna et al. (2025) propose determining the optimal threshold $\delta^\star$ as the value which sets the proportion of misclassification among pairs flagged as being 'non-null' at 50%. This is a False Discovery Rate (FDR), and implies that if $\delta^\star$ were lowered, more than half of the pairs added would actually be null, whereas if $\delta^\star$ were increased, more than half of the pairs removed would actually be non-null.

Empirically, the value $\delta^\star$ is determined with this trade-off in mind. In order to resolve this trade-off, we define as $F_0(\cdot)$ the (unknown) distribution function of empirical correlations $\widehat{\rho}_{i,i'}$ for **null** pairs of units (we discuss the estimation of this below). Similarly, we can define $F_0^+(\cdot) \equiv 2(1 - F_0(\cdot))$ as the right distribution for the *absolute* correlations $|\widehat{\rho}_{i,i'}|$.[3] Similarly, we denote as $F^+(\delta) = \Pr(|\widehat{\rho}_{i,i'}| \geq \delta)$ the right distribution function for absolute correlations for both null and non-null pairs of units (*i.e.* the unconditional distribution).

We can write the probability that observations whose $\widehat{\rho}_{i,i'} \geq \delta$ are non-null and null as follows for a specific threshold choice δ:

$$\Pr(\text{Non-Null}\big|\ |\widehat{\rho}_{i,i'}| \geq \delta) = \frac{F^+(\delta) - p_0 F_0^+(\delta)}{F^+(\delta)} \qquad \Pr(\text{Null}\big|\ |\widehat{\rho}_{i,i'}| \geq \delta) = \frac{p_0 F_0^+(\delta)}{F^+(\delta)}, \quad (8)$$

where p_0 refers to the proportion of pairs of units with a null relationship across space. We can then take the difference between the probability of observations being null and non-null as proportional to the following:

$$Q(\delta) = (F^+(\delta) - p_0 F_0^+(\delta)) - p_0 F_0^+(\delta) = F^+(\delta) - 2p_0 F_0^+(\delta). \quad (9)$$

A key property to note with $Q(\delta)$ is that this will be increasing in δ if more nulls are admitted than non-nulls, and decreasing in δ if more non-nulls are admitted than nulls. Thus, maximizing $Q(\delta)$ with respects to δ finds the point at which half the added pairs are null, and half are non-null.

This suggests an (infeasible) criteria to select delta as maximizing the following quantity $\widetilde{Q}$:

$$\widetilde{Q}_{N,d}(\delta) = \widehat{F}^+(\delta) - 2p_0 F_0^+(\delta) \qquad \text{where } \widehat{F}^+(\delta) = \frac{2}{N(N-1)} \sum_{i=1}^{N} \sum_{i'<i} \mathbb{1}\{|\widehat{\rho}_{i,i'}| > \delta\}. \quad (10)$$

This quantity cannot yet be estimated for two reasons. Firstly, while the distribution of both null and non-null pairs can be estimated from data as $\widehat{F}^+(\delta)$, the distribution of just null pairs $\widehat{F}_0^+(\delta)$ is unknown. And secondly, the proportion of nulls themselves (p_0) is

[3] Could point to an appendix where we simulate an example and show this in a step-by-step way so readers can easily understand the mechanics...?

also unknown. In line with the sparsity assumption, the quantity p_0 can (conservatively) be set equal to 1. This leads to a feasible criteria:

$$\widehat{Q}_{N,d}(\delta) = \widehat{F}^{+}(\delta) - 2F_0^{+}(\delta), \tag{11}$$

where the optimal threshold $\delta^\star$ is chosen as the quantity that maximizes (11).

Thus, a final step required to actually estimate $\widehat{Q}_{N,d}$ is to estimate the null distribution $F_0(\cdot)$. This cannot be estimated directly as the distribution of $\widehat{\rho}_{i,i'}$ given that this distribution will contain both null and non-null pairs. However, given that most pairs are assumed to actually be null pairs, this leads to the proposal in DellaVigna et al. (2025) to estimate this null distribution by fitting a Normal distribution to the central part of the empirical distribution of $\widehat{\rho}_{i,i'}$ which will overwhelmingly consist of null pairs. In the interests of completion, if $\Phi_v(\cdot)$ refers to the normal distribution with variance v, and $\Phi_v^{-1}(\cdot)$ its inverse, then the interquartile function can be written as $IQR_{v,0} = \Phi_v^{-1}(0.75) - \Phi_v^{-1}(0.25)$. Similarly, defining the empirical distribution of observed correlations as:

$$G_{N,d}(\delta) = \frac{2}{N(N-1)} \sum_{i=1}^{N} \sum_{i'<i} \mathbb{1}\{\widehat{\rho}_{i,i'} < \delta\},$$

we can define the inverse and interquartile range of observed correlations (respectively) as $G_{N,d}^{-1}(\cdot)$ and $\widehat{IQR} = G_{N,d}^{-1}(0.75) - G_{N,d}^{-1}(0.25)$. The distribution F_0^{+} is thus estimated by setting v to the quantity which resolves $IQR_{\widehat{v},0} = \widehat{IQR}$.

At the beginning of Section 4 we will examine an empirical example which steps through each of these steps, in essence considering the estimation of $\widehat{\rho}_{i,i'}$, $G_{N,d}(\delta)$, $\widehat{v}$, and $\delta^\star$, and resultingly, the standard errors and confidence intervals corresponding to our regression estimate of interest $\widehat{\tau}$. Examining this in practice helps to elucidate the steps in both thresholding, and estimating cross-space correlations. A summarised statement of the entire estimation algorithm as defined in DellaVigna et al. (2025) is provided as Algorithm 1 in an Appendix to this article.

2.2 Alternative Estimators

While the above treatment begins with a bivariate linear regression model in (1), the procedures extend directly to the estimation environments most commonly encountered in practice. With additional covariates or fixed effects, the residualised treatment $\tilde{W}_i$—the residual from regressing W_i on all other regressors—takes the place of W_i in the sandwich formula, by the Frisch–Waugh–Lovell theorem; the `tmo` command constructs $\tilde{W}_i$ internally from the command supplied in `cmd()`. With instrumental variables, the same logic applies to the first-stage projection: the command builds the appropriate 2SLS residualisation, so that the reported coefficient is the second-stage estimate while the variance is adjusted using TMO. In panel settings, residuals for each location are stacked across time periods, cross-location correlations are computed using all periods jointly, and the sandwich accumulates contributions over pairs of locations and pairs of periods. Formal treatments of each extension are provided in DellaVigna et al. (2025, Appendix B.3). In terms of implementation, `tmo` currently supports `regress`, `areg`, `reghdfe`, `ivregress`, `ivreg2` and `ivreghdfe`, as documented in Section 3.

2.3 Combining TMO with Distance-Based Procedures

TMO and the procedures reviewed above are not mutually exclusive. Each of the estimators discussed in this section can be written as a sandwich estimator in which the contribution of a pair of locations (i, i') enters with some weight $w_{ii'}$. Clustering sets $w_{ii'} = 1$ within clusters and 0 elsewhere; Conley-type estimators set $w_{ii'} = K(d_{ii'}/b)$, a kernel in distance; and SCPC implies a dense kernel $w_{ii'} = P_{ii'}$ derived from its low-frequency principal components, under which *every* pair receives some (typically small) weight.

This common structure makes combination natural. The combined estimator implemented in `tmo` retains the raw covariance term (weight 1) for every pair selected by the TMO threshold—and for every pair included by the baseline procedure, in the clustering and uniform-distance cases—while remaining pairs enter with the baseline weight:

$$\widehat{\mathbb{V}}_{combined} \propto \sum_{(i,i'):\,|\widehat{\rho}_{ii'}|\geq\widehat{\delta}} \widehat{\varepsilon}_i\widehat{\varepsilon}_{i'}\tilde{W}_i\tilde{W}_{i'} \;+\; \sum_{\text{remaining pairs}} w_{ii'}\,\widehat{\varepsilon}_i\widehat{\varepsilon}_{i'}\tilde{W}_i\tilde{W}_{i'}, \tag{12}$$

where the first summation reflects pairs flagged by TMO (as well as own-cluster or within-cutoff pairs, where relevant), and $w_{ii'}$ in the second summation is the baseline weight: $\mathbb{1}\{d_{ii'} \leq b\}$ for a uniform distance kernel, $\max(1 - d_{ii'}/b, 0)$ for a Bartlett kernel (the spatial analogue of Newey and West 1987), or the SCPC kernel weight $P_{ii'}$. The intuition for such combinations is that the two sources of information are complementary: distance-based procedures are well suited to diffuse, geographically organised dependence, while TMO captures strong dependencies between particular pairs of locations—which need not be geographically close—revealed by the auxiliary outcomes. Section 4 illustrates each combination, which is requested in `tmo` through the `cmd()` (clustering), `distthreshold()/distkernel()` (distance), and `scpc_cmd()` (SCPC) options.

3 The `tmo` command

3.1 Syntax

The `tmo` command is an e-class command with the following syntax:

`tmo, cmd(`*cmdline*`) x(`*varname*`) ylist(`*varlist*`) idvar(`*varname*`)` [`options`]

3.2 Options

`cmd(`*cmdline*`)` *cmdline* is the command that produces the regression of interest. tmo currently supports regressions using `regress`, `areg`, `reghdfe`, `ivregress`, `ivreg2`, or `ivreghdfe`.

`x(`*varname*`)` Regressor of interest in `cmd` for which to estimate TMO standard errors. tmo *estimates the standard error for only this declared independent variable.*

`ylist(`*varlist*`)` *List of auxiliary outcomes to use in* tmo.

`idvar(`*varname*`)` Location identifier variable; must be unique (within each time period for panel case).

`misslimit(`#`)` Limit for proportion of observations allowed to be missing for auxiliary outcomes. Auxiliary outcomes missing more than `misslimit` are not used. Must be between [0,1]; default is 0.1.

Panel Setting

`timevar(`*varname*`)` Time identifier variable; must be declared for panel case.

Distance-based Settings

`latitude(`*varname*`)` Latitude variable in signed decimal degrees.

`longitude(`*varname*`)` Longitude variable in signed decimal degrees.

`distthreshold(`#`)` Distance threshold to allow for arbitrary correlation between pairs of locations that are `distthreshold` or closer. Distances are measured in kilometres by default; see the `miles` option. Combines tmo *with a Conley adjustment using a uniform kernel (but see* `distkernel()`*). Requires* `latitude` and `longitude`.

`miles Measure distances for distthreshold() in miles rather than kilometres.`

`distkernel(`*str*`)` `Kernel for the distance-based adjustment; either uniform (the default: pairs within distthreshold() receive full weight) or bartlett (pairs not selected by the TMO threshold receive weight` $1-d_{ii'}/b$`, declining linearly to zero at the cutoff` b`, analogous to the pair-level SCPC combination). Requires distthreshold(); may not be combined with scpc_cmd().`

Saving Figures and Tables

`filesuffix(`*str*`) Folder path and base filename for saving figures and results. Required for plot` or `save` options.

`savedyad` Save Stata data file with correlation and contribution to standard error for each location pair.

`plotq` Save plot of optimal threshold estimator.

`plothist` Save plot for histogram of correlations between locations.

`plothistnbins(`#`)` Number of bins for histogram of correlations (default 10000).

`plotse` Save plot for standard error estimates across thresholds.

`saveplotseest` Save Stata data file with standard error estimates across thresholds.

`saveest` Save results in `r()` to Stata data file.

Custom Threshold

`threshold(`#`)` Set custom threshold instead of using the optimal threshold from the interquartile range method. Must be between [0,1].

`thresholdoff` Turns off the tmo *adjustment entirely.*

SCPC Options

`scpc_cmd (`*cmdline*`)` Command for regression of interest before applying SCPC adjustment. Combines tmo with the SCPC method from Müller and Watson (2022b): pairs selected by the TMO threshold enter the variance with their raw covariance, while all remaining pairs enter through the implicit SCPC kernel weight. Requires `latitude` and `longitude`.

`scpc_uncond` Turns on the unconditional SCPC inference setting.

Returned Objects

`tmo` is an e-class command, and stores the following in `e()`:

Scalars:

`e(beta)`	point estimate for the variable declared in `x()`
`e(orig_se)`	standard error from the original command in `cmd()`
`e(tmo_se)`	TMO-adjusted standard error
`e(lb)`, `e(ub)`	bounds of the TMO-adjusted 95% confidence interval
`e(threshold)`	optimal (or user-set) threshold $\hat{\delta}$
`e(pct_ge_thres)`	% of off-diagonal pairs included in the variance
`e(pct_ge_thres_nocl)`	% of pairs over the threshold outside clusters/Conley
`e(N)`, `e(N_loc)`, `e(N_clust)`	observations, locations, and clusters
`e(T)`	time periods (panel case)
`e(N_outcomes)`	number of auxiliary outcomes used
`e(dof)`	degrees of freedom of the fitted null distribution
`e(finite_sample_dof)`	finite-sample degrees-of-freedom adjustment
`e(df_r)`	residual degrees of freedom
`e(r2)`, `e(r2_a)`, `e(rmse)`	fit statistics from the original command
`e(scpc_cv)`	SCPC critical value (when `scpc_cmd()` is used)

Matrices:

The matrices `e(b)` and `e(V)` contain the point estimate and TMO-adjusted variance for the variable declared in `x()`, facilitating the exportation of results with routines such as `estout`.

4 Examples

4.1 An Illustrative Example: OLS and US counties

It is illustrative to view the principles of TMO in a specific empirical setting. To do so we will consider an example from DellaVigna et al. (2025) in which a rich collection of county-level outcomes for the United States has been collated, principally from U.S. Census sources. The dataset (`county_differences.dta`, bundled with the command's repository) contains, for each of roughly 3,100 counties, long-differences of over 90 standardised economic and social outcomes: income, employment by sector, education, health, and demographic measures, among others. Each variable is expressed as the change over 1980–2009 and normalised to mean zero and unit variance. We estimate the effect of the change in the share of college graduates (`EDU_college_d`) on the change in per-capita personal income (`PIN_persincpc_d`), including state fixed effects. The county FIPS code (`fips`) identifies locations, and every other outcome in the dataset serves as an auxiliary outcome in `ylist()`:

```
. tmo, cmd(reg PIN_persincpc_d EDU_college_d i.stfips, r) x(EDU_college_d) ///
>     ylist(`ylist´) i(fips) plothist plotq file("figures/")
file figures/_qt.pdf saved as PDF format
file figures/_hist.png written in PNG format

Linear Regression with TMO                             Number of obs =  3,060
                                                       R-squared     = 0.3434
                                                       Adj R-squared = 0.3434
```

Root MSE = 0.1409

PIN_persin~d	Coefficient	Std. err.	t	P>|t|	[95% conf. interval]	
EDU_colleg~d	.1745893	.0197081	8.86	0.000	.1359466	.213232

Optimal threshold	0.389
% of off-diag in SE est.	0.497
% >= threshold (excl. clusters/Conley)	0.497
# outcomes	90.000
Degrees of freedom	59.986

The upper panel reports the conventional OLS results. The lower block summarizes the TMO adjustment. The *Optimal threshold* $\hat{\delta}$ is a data–driven cutoff for the absolute correlations across locations; pairs with $|\hat{\rho}_{i,i'}| \geq \hat{\delta}$ are treated as correlated when constructing $\widehat{\Sigma}^{(0)}(\hat{\delta})$. The *% of off-diag in SE est.* row reports the share of off–diagonal entries retained by this rule—here roughly 0.5% of all county pairs, underscoring the sparsity of the selection—and thus the extent of cross–location dependence incorporated in the variance. When TMO augments clustering or Conley, this row also records the additional pairs contributed by the baseline method. In this case 90 auxiliary outcomes were used to learn the correlation structure; finally, the degrees of freedom reports the width of the normal reference used to set $\hat{\delta}$ (estimated from the interquartile range of Fisher–z correlations).

The figures document the choice of $\hat{\delta}$. Figure 2 shows the empirical distribution of Fisher–z correlations across locations together with the fitted $N(0, \hat{v})$ reference; the dashed vertical lines at $\pm\hat{\delta}$ indicate which pairs populate the off–diagonal of $\widehat{\Sigma}^{(0)}(\hat{\delta})$. Figure 3 then plots the threshold objective $\hat{Q}_{n,d}(\delta)$; the dashed line marks its maximizer at $\hat{\delta}$ (larger δ yields a more conservative selection of pairs). The reported standard error is computed as $(W^\top W)^{-1}\big(W^\top \widehat{\Sigma}^{(0)}(\hat{\delta})W\big)(W^\top W)^{-1}$.

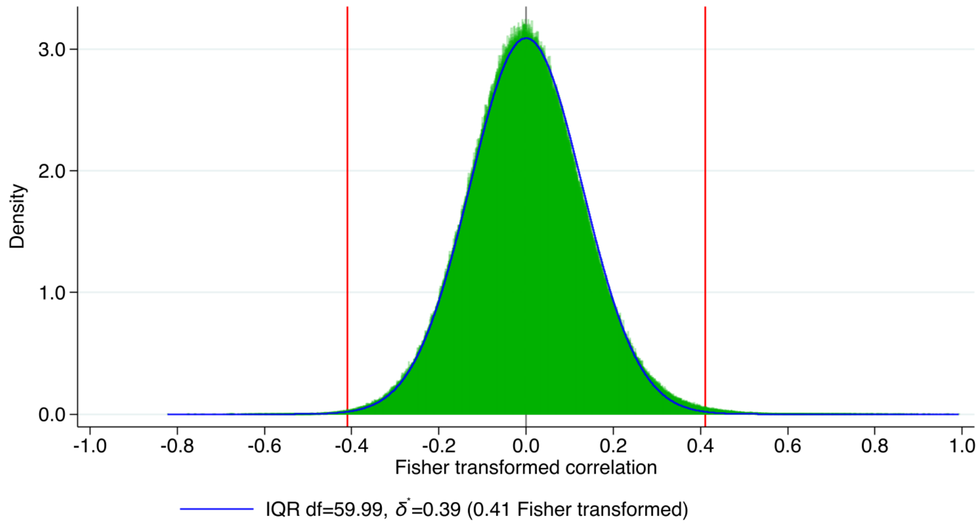


Figure 2: Correlations between residuals across US counties

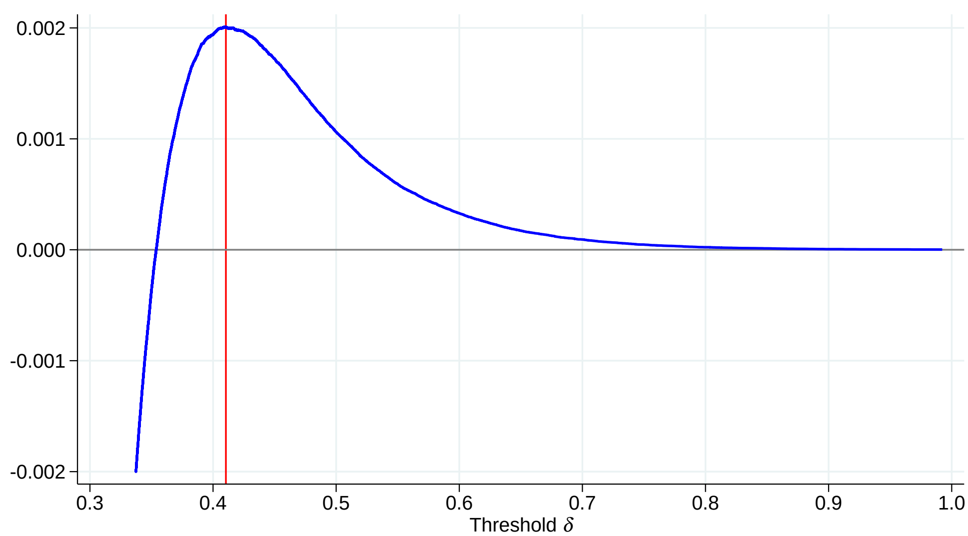


Figure 3: Optimal Thresholding

4.2 Alternative Models

Instrumental Variables `tmo` can be used with instrumental–variables estimators. Using the same cross sectional dataset, we estimate the effect of infant mortality on life expectancy instrumenting `VST_infmort_d` with the median distance (in miles) to the nearest emergency department, obstetrics department, and pediatric ICU. The `cmd()` option simply wraps the IV estimator you would normally use; its contents follow the *native* syntax of the chosen command. `tmo` is compatible with `ivreg2`, `ivreghdfe`, and `ivregress`. The option `x()` marks the endogenous/treatment variable whose coefficient's variance is to be adjusted, and `ylist()` supplies the auxiliary outcomes.

```
. tmo, cmd(ivreg2 life_d (VST_infmort_d = AHRQ_emerdist_d AHRQ_obgyndist_d ///
>     AHRQ_pediadist_d)) x(VST_infmort_d) ylist(`ylist´) i(fips)

Linear Regression with TMO                          Number of obs =   3,060
                                                    R-squared     = -0.0464
                                                    Adj R-squared = -0.0464
                                                    Root MSE      =  0.0460

------------------------------------------------------------------------------
      life_d | Coefficient  Std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
VST_infmor~d |  -.0082797    .015985    -0.52   0.604    -.0396097    .0230503
------------------------------------------------------------------------------
-------------------------------------------------------------------------------

                                              Optimal threshold      0.502
                                          % of off-diag in SE est.   5.843
                         % >= threshold (excl. clusters/Conley)      5.843
                                                     # outcomes     87.000
                                              Degrees of freedom    22.522
-------------------------------------------------------------------------------
```

The output mirrors the cross sectional case: the reported coefficient equals the second–stage IV estimate from the underlying command, while the standard error is TMO–adjusted; the lower block reports the optimal threshold, the share of pairs retained, and the remaining diagnostic quantities, all with the same interpretation as before.

Panel Data Panel–ready data are available as `county_panel.dta` in the project repository on GitHub.

In panel settings, `tmo` incorporates the time dimension through the unit and time identifiers specified in `idvar()` and `timevar()`. The command treats $\{i, t\}$ as a panel, reshapes the normalized residuals into an $N \times T$ array, and computes cross location correlations using all periods. The sandwich matrix is then built from location pairs (i, i') combined with time pairs, which increases computational cost; for large panels the program prints occasional progress messages ("`Computed X% of sandwich`") as it accumulates contributions by blocks.

```
. tmo, cmd(reg EMN_farm EDU_publicenroll i.year i.stfips, cluster(fips)) ///
>     x(EDU_publicenroll) ylist(`ylist´) i(fips) t(year)
```

```
Linear Regression with TMO                               Number of obs =  6,222
                                                         R-squared     = 0.3426
                                                         Adj R-squared = 0.3426
                                                         Root MSE      = 0.7683

------------------------------------------------------------------------------
    EMN_farm | Coefficient  Std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
EDU_public~l |   .1355818   .0758575     1.79   0.074    -.0131541    .2843177
------------------------------------------------------------------------------
-------------------------------------------------------------------------------

                                               Optimal threshold      0.545
                                         % of off-diag in SE est.      0.892
                         % >= threshold (excl. clusters/Conley)      0.892
                                                      # outcomes     63.000
                                              Degrees of freedom     24.612
-------------------------------------------------------------------------------
```

The output is analogous to the cross sectional case. The effective sample size increases because the data expand over time, and the optimal threshold may differ since correlations are constructed using the panel information. All reported fields keep the same interpretation as in the cross sectional setting.

Other routines The regression of interest may be specified with any of the supported estimation commands, and equivalent specifications yield identical results regardless of the wrapper used. The log below estimates the same state-fixed-effects model three ways—`regress` with factor-variable notation, `reghdfe` with `absorb()`, and `areg`—and reports timings. The three TMO standard errors coincide; run-times differ only through the speed of the underlying estimation command, since the auxiliary-outcome regressions are re-estimated with the same wrapper.

```
. timer clear
. timer on 1
. qui tmo, cmd(regress PIN_persincpc_d EDU_college_d i.stfips, r) ///
>     x(EDU_college_d) ylist(`ylist´) i(fips)
. timer off 1
.
. timer on 2
. qui tmo, cmd(reghdfe PIN_persincpc_d EDU_college_d, vce(r) abs(stfips)) ///
>     x(EDU_college_d) ylist(`ylist´) i(fips)
. timer off 2
.
. timer on 3
. qui tmo, cmd(areg PIN_persincpc_d EDU_college_d, r abs(stfips)) ///
>     x(EDU_college_d) ylist(`ylist´) i(fips)
. timer off 3
. timer list
   1:      6.00 /        1 =       6.0040
   2:      6.03 /        1 =       6.0300
   3:      5.87 /        1 =       5.8730
```

4.3 Alternative settings

Combining `tmo` with other spatial dependence procedures

The `tmo` command can be combined with existing approaches to spatial inference. This is useful when the researcher wants to preserve a conventional restriction on the covariance matrix—for example arbitrary dependence within states, within a distance cutoff, or through the SCPC kernel—while allowing TMO to add further pairs whose auxiliary outcomes suggest residual dependence. Operationally, the original procedure defines a baseline set of pairs that enter the variance estimator. TMO then adds pairs with $|\widehat{\rho}_{ii'}| \geq \widehat{\delta}$, except where the baseline procedure already includes them.

TMO with clustered standard errors The simplest combination starts from the usual cluster-robust estimator. In the example below, the original regression clusters by state, and `tmo` is then called with the same clustered regression in `cmd()`. The command preserves the within-state covariance terms and uses the auxiliary outcomes to decide which cross-state pairs should also enter the variance.

```
. reg PIN_persincpc_d EDU_college_d, cluster(stfips)

Linear regression                               Number of obs     =      3,060
                                                F(1, 47)          =      39.74
                                                Prob > F          =     0.0000
                                                R-squared         =     0.0708
                                                Root MSE          =     .16767

                                (Std. err. adjusted for 48 clusters in stfips)
------------------------------------------------------------------------------
             |               Robust
PIN_persin~d | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
EDU_colleg~d |   .2299788   .0364834     6.30   0.000     .1565835     .303374
       _cons |   1.934663   .0243343    79.50   0.000     1.885708    1.983617
------------------------------------------------------------------------------

. tmo, cmd(reg PIN_persincpc_d EDU_college_d, cluster(stfips)) ///
>     x(EDU_college_d) ylist(`ylist´) i(fips)

Linear Regression with TMO                              Number of obs =  3,060
                                                        R-squared     = 0.0705
                                                        Adj R-squared = 0.0705
                                                        Root MSE      = 0.1677

------------------------------------------------------------------------------
PIN_persin~d | Coefficient  Std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
EDU_colleg~d |   .2299788   .0381001     6.04   0.000     .1533312    .3066263
------------------------------------------------------------------------------
-------------------------------------------------------------------------------

                                         Optimal threshold      0.433
                                    % of off-diag in SE est.      3.508
                           % >= threshold (excl. clusters/Conley)      0.438
```

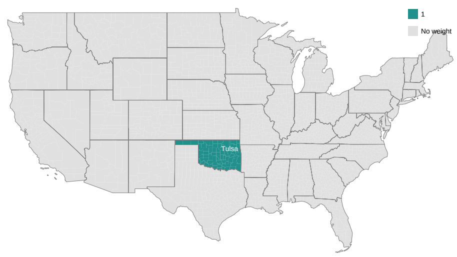
```
                                              # outcomes     90.000
                                      Degrees of freedom     45.192
```

TMO with distance-based adjustments Distance-based adjustments are requested by supplying county centroids through `latitude()` and `longitude()`, together with a distance cutoff through `distthreshold()`. The default kernel is uniform, so every pair inside the cutoff enters with full weight. The development version also provides `distkernel(bartlett)`, which assigns a linearly declining weight to non-TMO-selected pairs inside the cutoff while retaining TMO-selected pairs with full weight.

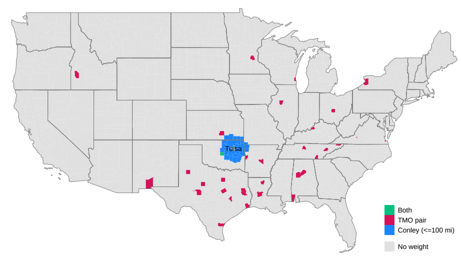
```
. tmo, cmd(reg PIN_persincpc_d EDU_college_d, r) ///
>     x(EDU_college_d) ylist(`ylist´) i(fips) lat(_CY) lon(_CX) ///
>     distthreshold(100) miles
Linear Regression with TMO                             Number of obs =  3,060
                                                       R-squared     = 0.0705
                                                       Adj R-squared = 0.0705
                                                       Root MSE      = 0.1677

PIN_persin~d | Coefficient  Std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
EDU_colleg~d |   .2299788    .035119     6.55   0.000     .1611196    .2988379
------------------------------------------------------------------------------

                                           Optimal threshold      0.433
                                    % of off-diag in SE est.      1.943
                      % >= threshold (excl. clusters/Conley)      0.492
                                                  # outcomes     90.000
                                          Degrees of freedom     45.192

. tmo, cmd(reg PIN_persincpc_d EDU_college_d, r) ///
>     x(EDU_college_d) ylist(`ylist´) i(fips) lat(_CY) lon(_CX) ///
>     distthreshold(100) miles distkernel(bartlett)
Linear Regression with TMO                             Number of obs =  3,060
                                                       R-squared     = 0.0705
                                                       Adj R-squared = 0.0705
                                                       Root MSE      = 0.1677

PIN_persin~d | Coefficient  Std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
EDU_colleg~d |   .2299788    .029723     7.74   0.000     .1716996    .2882579
------------------------------------------------------------------------------

                                           Optimal threshold      0.433
                                    % of off-diag in SE est.      1.943
                      % >= threshold (excl. clusters/Conley)      0.492
                                                  # outcomes     90.000
                                          Degrees of freedom     45.192
```

TMO with SCPC The SCPC combination is requested with `scpc_cmd()`. In this case, TMO first runs the SCPC regression using latitude and longitude ordered as `s_1` and `s_2`, respectively. Pairs selected by TMO enter with the raw residual covariance, while all remaining pairs enter through the implicit SCPC kernel weight. This gives a direct implementation of the combined TMO–SCPC estimator in the same syntax used for the other examples.

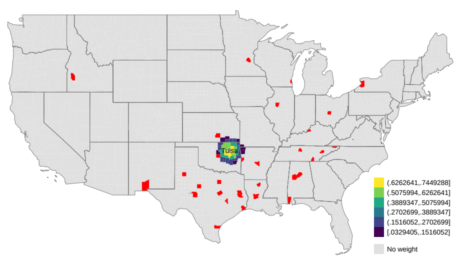
```
. reg PIN_persincpc_d EDU_college_d, r

Linear regression                               Number of obs     =      3,060
                                                F(1, 3058)        =     168.81
                                                Prob > F          =     0.0000
                                                R-squared         =     0.0708
                                                Root MSE          =     .16767

------------------------------------------------------------------------------
             |               Robust
PIN_persin~d | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
EDU_colleg~d |   .2299788   .0177005    12.99   0.000     .1952727    .2646848
       _cons |   1.934663   .0089404   216.40   0.000     1.917133    1.952192
------------------------------------------------------------------------------

. scpc, latlong
found 3060 observations / clusters and 2-dimensional locations in s_*
Computing distances on surface of sphere treating s_1 as latitude and s_2 as lo
> ngitude
SCPC optimal q = 6 for maximal average pairwise correlation = 0.030

SCPC Inference for first 2 coefficients
              |        Coef     Std_Err       t      P>|t|    95% Conf
--------------+----------------------------------------------------------
 EDU_colleg~d |    .2299788    .0253915     9.06     0.000    .1625168
        _cons |    1.934663    .0335045    57.74     0.000    1.843191
              |    Interval
--------------+-------------
 EDU_colleg~d |    .2974407
        _cons |    2.026134
. rename (s_1 s_2) (_CY _CX)
.
. tmo, cmd(regress PIN_persincpc_d EDU_college_d, r) ///
>     x(EDU_college_d) ylist(`ylist´) i(fips) lat(_CY) lon(_CX) ///
>     scpc_cmd(reg PIN_persincpc_d EDU_college_d, r)

Linear Regression with TMO                              Number of obs =  3,060
                                                        R-squared     = 0.0705
                                                        Adj R-squared = 0.0705
                                                        Root MSE      = 0.1677

------------------------------------------------------------------------------
PIN_persin~d | Coefficient  Std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
EDU_colleg~d |   .2299788   .0287255     8.01   0.000     .1736556    .2863019
------------------------------------------------------------------------------
-------------------------------------------------------------------------------

                                             Optimal threshold      0.433
                                      % of off-diag in SE est.      0.667
                         % >= threshold (excl. clusters/Conley)     0.667
                                                    # outcomes     90.000
                                            Degrees of freedom     45.192
```



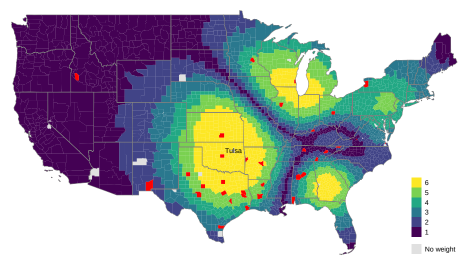
(a) State clustering



(b) TMO + Conley



(c) TMO + Bartlett distance kernel



(d) TMO + SCPC

Figure 4: Combining TMO with existing spatial-dependence adjustments

Figure 4 visualises each combination for the focal county of Figure 1, with every layer derived from the pair-level output of the corresponding `tmo` run. Two features deserve emphasis, since the combined estimators inherit very different structures from their two components. First, the TMO contribution is deliberately *sparse*: only the small set of counties whose auxiliary outcomes systematically co-move with the focal county is selected (shown in red in panels c and d), and these counties may lie anywhere on the map. Second, the baseline contributions differ in their density. The Conley disk (panel b) and Bartlett kernel (panel c) are *local and bounded*: weights are confined within the distance cutoff, and are zero everywhere else—so that the map is mostly unshaded. The SCPC kernel (panel d, background), by contrast, is *dense and global*: because SCPC conducts inference through a handful of smooth low-frequency spatial components, every pair of counties receives some weight, without any threshold or cutoff, and the entire map is therefore shaded. This is not an artefact of the figure but a faithful depiction of the estimator: in the combined TMO–SCPC variance, the red counties contribute their raw residual covariance, while every remaining county contributes through the smoothly varying kernel weight shown in the background. The comparison also clarifies the division of labour in equation (12): the baseline captures diffuse, geographically organised dependence—sharply bounded for Conley, smooth and global for SCPC—while TMO adds the specific, potentially distant, pairs that geography alone would miss.

When should you use `tmo`? A diagnostic guide, applied to Acemoglu et al. (2019)

The examples so far document *how* to run the command. The more important question for a practitioner is *when*: given a particular dataset and regression, is TMO needed at all, and should it replace or complement a distance-based correction? DellaVigna et al. (2025) propose a series of empirical diagnostics to answer precisely these questions, and in this subsection we walk through them, end to end, in a well-known cross-country application. Acemoglu et al. (2019) estimate the effect of democratisation on economic growth in a panel of 175 countries observed between 1960 and 2010, in the dynamic specification

$$\log y_{ct} = \beta\, dem_{ct} + \sum_{j=1}^{4} \gamma_j \log y_{c,t-j} + \alpha_c + \delta_t + \epsilon_{ct},$$

with standard errors clustered by country (their Table 2, Column 3, described by the authors as their preferred specification). Reproducing this specification with `areg`—the within estimator, identical to the original `xtreg, fe`—returns the published point estimate exactly, $\widehat{\beta} = 0.787$, with a cluster-robust standard error of 0.230 (0.226 under `xtreg`'s degrees-of-freedom convention).[4] As auxiliary outcomes we assemble roughly thirty country-year economic series from the replication package—education attainment and enrolment, trade, investment, TFP, child mortality, tax revenues, market reforms, unrest, population structure, and external assets—excluding the package's alternative democracy indices, which measure the treatment rather than outcomes. The log below runs `tmo` augmenting the original clustering, and requests the two diagnostic figures discussed next:

```
. tmo, cmd(areg y dem ly1 ly2 ly3 ly4 yy*, absorb(wbcode2) cluster(wbcode2)) //
> /
>     x(dem) ylist(`ylist´) i(wbcode2) t(year) ///
>     plothist plothistnbins(100) plotse savedyad file("figures/acemoglu")
(file figures/acemoglu_dyad.dta not found)
file figures/acemoglu_dyad.dta saved
file figures/acemoglu_hist.png written in PNG format
Calculating TMO SE over threshold at 0.00
Calculating TMO SE over threshold at 0.20
Calculating TMO SE over threshold at 0.40
Calculating TMO SE over threshold at 0.60
Calculating TMO SE over threshold at 0.80
Calculating TMO SE over threshold at 1.00
file figures/acemoglu_se_ratio_over_thres.pdf saved as PDF format
```

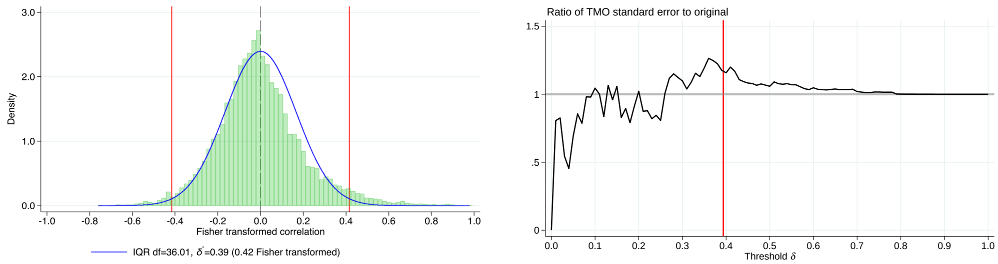
[4] Two practical notes for dynamic panels: lags should be generated as ordinary variables before calling `tmo`, rather than written with time-series operators inside `cmd()` (the command subsets internally to the estimation sample, and in an unbalanced panel time-series operators would silently re-compute lags across gaps), and the regressor of interest should be listed immediately after the dependent variable in `cmd()`.

```
file figures/acemoglu_prop_above_thres.pdf saved as PDF format
Linear Regression with TMO                             Number of obs =  6,336
                                                       R-squared     = 0.9989
                                                       Adj R-squared = 0.9989
                                                       Root MSE      = 5.0279
```

y	Coefficient	Std. err.	t	P>|t|	[95% conf. interval]	
dem	.7865534	.2683049	2.93	0.004	.2570023	1.316105

Optimal threshold	0.394
% of off-diag in SE est.	5.399
% >= threshold (excl. clusters/Conley)	5.399
# outcomes	32.000
Degrees of freedom	36.011

Diagnostic 1: is there dependence for TMO to find, and can it be measured reliably? DellaVigna et al. (2025) recommend two checks before trusting any TMO adjustment, both produced by the command (`plothist` and `plotse`) and displayed in Figure 5. The figures are not decoration: each comes with a specific list of features to verify. In the correlation histogram (panel a), check that (i) the *centre* of the distribution is symmetric, unimodal, and closely tracked by the fitted Gaussian null—the central mass is dominated by null pairs, and a skewed or poorly-fitted centre indicates too much common correlation across the auxiliary outcomes, invalidating the estimated null; (ii) there is visible *excess mass in the tails* relative to the fitted null—that excess *is* the evidence of non-null pairs which TMO exists to capture, and without it there is nothing for the procedure to find; and (iii) the implied degrees of freedom ($\widehat{df} = 1/\hat{v}$, reported in the output) reach at least 20, the minimum DellaVigna et al. recommend, with more advisable in panel settings. Here the fit is adequate, the tail excess is clear, and $\widehat{df} = 36$. In the threshold plot (panel b), check that (i) the curve is *smooth and forms a plateau* around $\hat{\delta}^{\star}$, so that the reported adjustment does not hinge on the exact threshold; (ii) the ratio tends to one as $\delta \to 1$ (with no pairs selected the baseline standard error must be recovered—a mechanical sanity check); and (iii) erratic movement is confined to low thresholds, where selection is dominated by noise pairs. With only 175 locations the left portion of our curve is visibly noisier than in county-level applications—exactly the small-N caution raised by DellaVigna et al., who show that with as few as 60 countries (*e.g.*, in their re-analysis of Funke et al. 2023) the histogram centre is visibly non-Gaussian, the threshold curve is erratic, and the procedure is unreliable—but it is stable around the optimum $\hat{\delta}^{\star} = 0.39$. When either diagnostic fails, the remedy is a larger or more relevant collection of outcomes, not a different threshold.



(a) Correlations and fitted null (b) SE ratio across thresholds

Notes: produced by the `plothist` and `plotse` options. The panels are the direct analogues of Figures E.7 and E.8 in DellaVigna et al. (2025) for this application, and display the same qualitative anatomy (Gaussian-tracked centre with tail excess; noisy left segment, plateau at the optimum, convergence to one). The quantitative differences—$\widehat{df} = 36$ and $\hat{\delta}^{\star} = 0.39$ here, versus 79 and 0.27 in the published figures—are entirely attributable to the auxiliary collections (the package's 32 outcomes versus their externally supplemented 79), as documented in Table 3.

Figure 5: Diagnostics for the application to Acemoglu et al. (2019)

Diagnostic 2: is the dependence geographic? This is the question that decides between corrections. Following Table 1 of DellaVigna et al. (2025), Table 1 asks what the country pairs selected by TMO ($|\hat{\rho}_{ii'}| \geq \hat{\delta}^{\star}$, some 800 pairs) have in common. Geographic proximity is informative but startlingly incomplete: only 11% of selected pairs lie within 650 miles of one another (the Conley bandwidth DellaVigna et al. adopt at the country level). Being in the same world region does far better (47%), initial income proximity adds little, and even taking all three criteria together leaves *half* of the selected pairs unexplained. The flavour of the unexplained half is instructive: among the most strongly correlated country pairs sit Sweden–Finland (317 miles apart), but also the United States–United Kingdom ($\hat{\rho} = 0.83$ at 4,400 miles) and Indonesia–Colombia ($\hat{\rho} = 0.67$ at 11,800 miles)—pairs linked by common growth trajectories, trade exposure, or commodity dependence rather than by geography, and invisible to any distance kernel.

	Among pairs with $\lvert\hat{\rho}_{ii'}\rvert \geq \hat{\delta}^{\star}$	Among all pairs
Within 650 miles	10.7%	2.9%
Same world region	46.7%	16.7%
Top 10% closest, 1960 GDP pc	16.4%	10.0%
Any of the above	51.7%	21.5%

Notes: share of country pairs satisfying each proximity criterion, among the 822 pairs selected by the TMO threshold ($\hat{\delta}^{\star} = 0.39$) and among all country pairs. Computed from the pair-level file saved by `savedyad`.

Table 1: Predictors of highly-correlated country pairs, Acemoglu et al. (2019) data

Diagnostic 3: how much does each correction move the standard error? Table 2 completes the argument by comparing the corrections directly, alongside the corresponding estimates in DellaVigna et al. (2025, Table 5). The Conley adjustment barely moves the standard error (a ratio of 1.03—identical, along with its share of admitted pairs, to the published figure, a useful validation of the implementation): consistent with Diagnostic 2, the dependence in these data is simply not organised by distance. TMO, by contrast, raises the standard error by 17%, and adding the Conley disk on top of TMO changes essentially nothing—the distance-based pairs contribute little that the auxiliary outcomes had not already found. Our adjustment is larger than the published 1.06, which is based on a collection of 79 auxiliary outcomes (the package's series supplemented with UN and World Bank data); the contrast is a live illustration of DellaVigna et al.'s recommendation that the collection of outcomes—their number *and* their relevance—is the researcher's most consequential choice. In all cases the effect of democratisation remains statistically significant.

	SE	Ratio to clustered	DellaVigna et al. (2025)
Cluster by country	0.230	1.00	1.00
Conley (650 mi)	0.236	1.03 [2.9%]	1.03 [2.9%]
TMO	0.268	1.17 [5.4%]	1.06 [6.0%]
TMO + Conley (650 mi)	0.268	1.17 [7.8%]	1.07 [8.1%]
SCPC / TMO + SCPC		—	1.01 / 1.10

Notes: ratios relative to the country-clustered standard error (0.230); brackets report the share of off-diagonal country pairs admitted to the variance. The final column reports the corresponding ratios in DellaVigna et al. (2025, Table 5), whose TMO figures use 79 auxiliary outcomes. The SCPC rows are omitted for our implementation: the current `scpc` release does not support this clustered dynamic panel through `scpc_cmd()`.

Table 2: Method comparison, Acemoglu et al. (2019) application

Diagnostic 4: how sensitive is the adjustment to the outcome collection? The choice of auxiliary outcomes is the researcher's most consequential decision, and both directions of error are possible. DellaVigna et al. (2025) show, in their re-analysis of Bernini et al. (2023), that the adjustment *grows* with the number of relevant outcomes—with few outcomes, correlations are noisily estimated, the null distribution widens, the optimal threshold rises, and true non-null pairs are missed—but also that *indiscriminately adding* outcomes can hurt: appending some sixty less-relevant external series to their collection *diluted* the estimated adjustment, because outcomes that do not share the correlation structure of the main outcome drag the correlation estimates toward zero. More is therefore not automatically better; the degrees-of-freedom statistic—which measures the *effective* number of independent outcomes, not the raw count—is the metric to watch, together with the stability of the adjustment across reasonable collections. Table 3 conducts this sensitivity exercise in our application. With a thin one-series-per-concept collection (18 outcomes), $\widehat{df}$ sits at the recommended floor of 20, the threshold jumps to 0.50, and the "adjustment" lands *below* one—an unstable, noise-dominated selection, not evidence that clustering was conservative. The full package collection (32 outcomes, $\widehat{df} = 36$) delivers the stable 1.17. DellaVigna et al.'s supplemented 79-outcome collection ($\widehat{df} = 79$) continues the pattern: a lower threshold, a sharper selection, and a moderated ratio of 1.06. The swing from 0.95 to 1.17 across nested, defensible collections is itself the diagnostic: had we stopped at the thin list, we would have reported essentially no correction, for lack of statistical power rather than lack of dependence.

Auxiliary collection	d	$\widehat{df}$	$\hat{\delta}^{\star}$	SE ratio
Package, one series per concept	18	20.8	0.50	0.95
Package, all economic series (baseline)	32	36.0	0.39	1.17
Package + treatment measures (misuse)	39	56.5	0.33	1.15
DellaVigna et al. (2025), supplemented	79	79.1	0.27	1.06

Notes: TMO applied to the Acemoglu et al. (2019) specification under alternative auxiliary collections. d is the number of outcomes used, $\widehat{df}$ the estimated degrees of freedom of the null, and the SE ratio is relative to the country-clustered standard error. The third row deliberately adds seven alternative democracy measures to illustrate a misuse that the diagnostics do not flag (see text). The final row reports the published figures.

Table 3: Sensitivity of the TMO adjustment to the auxiliary outcome collection

When badly-implemented TMO misleads. We close by cataloguing the failure modes a user should anticipate—several of which, importantly, are *not* detectable from the diagnostic figures, and must instead be ruled out by design. First, **including measurements of the treatment among the auxiliary outcomes**. The third row of Table 3 adds the package's seven alternative democracy indices to the collection: every reported diagnostic *improves* ($\widehat{df}$ rises to 56, the threshold falls), and the ratio looks unremarkable—yet the design is now invalid, since the "dependence" recovered from treatment-correlated outcomes partly reflects the spatial pattern of the treatment itself, biasing the adjustment in an unknown direction. The lesson is that outcome selection is an identification choice, not a diagnostic one. Second, **operating at the degrees-of-freedom floor**: as the first row shows, an underpowered collection produces thresholds and adjustments that swing widely—including below one—across small perturbations of the outcome list. Researchers should report the sensitivity exercise of Table 3, not a single run. Third, **too few locations**: with samples in the low hundreds the null approximation itself degrades, and below roughly a hundred locations (as in DellaVigna et al.'s 60-country re-analysis of Funke et al. 2023) both diagnostic figures visibly fail. Fourth, **unprepared outcome data**: the auxiliary series should be cleaned to a common scale before use—DellaVigna et al. (2025, Appendix B.1) normalise size-dependent variables by population, take logarithms of positive levels, standardise within period, and winsorise extreme tails—since a single heavy-tailed outcome can otherwise dominate the correlation estimates. Fifth, **pervasive spatial dependence**: TMO's sparsity assumption requires that most pairs be (nearly) uncorrelated; in data with strong global trends of the spatial-unit-root type, no threshold separates null from non-null pairs, the histogram centre fails the Gaussian check, and trend removal must precede any correction.

Taking stock. The pattern of diagnostics maps directly into advice. When the correlation-distribution and threshold-stability checks pass, and Table 1-style diagnostics show dependence concentrated at short distances, distance-based corrections are appropriate and TMO should confirm rather than change them. When—as here—a substantial share of detected dependence lies beyond any plausible distance kernel, distance-based corrections understate uncertainty and TMO (alone or combined) is called for. And when the diagnostics fail—few locations, a thin or irrelevant outcome collection, degrees of freedom at or below roughly 20, or instability of the adjustment across reasonable outcome sets—the honest conclusion is that TMO cannot be applied reliably, and enriching the auxiliary collection comes before any standard-error claim.

5 Discussion and Conclusion

In this article we have reviewed the main procedures available to Stata users for conducting inference in the presence of spatial dependence, and introduced the `tmo` command, which implements the Thresholding Multiple Outcomes procedure of DellaVigna et al. (2025). Rather than positing where spatial correlation lies—inside administrative borders, or within a chosen distance—TMO estimates the location of dependence from auxiliary outcomes observed for the same units, and then permits arbitrary correlation among the (typically very small) set of location pairs whose outcomes systematically co-move.

Several practical lessons emerge from the examples documented here. First, the procedure is undemanding in terms of syntax: a researcher who can estimate their model with `regress`, `areg`, `reghdfe`, `ivregress`, `ivreg2` or `ivreghdfe` can obtain TMO-adjusted standard errors by wrapping the same command line in `cmd()`. Second, the quality of the adjustment rests on the auxiliary outcomes: the proportionality assumption requires that the available outcomes span the relevant dimensions of cross-location dependence, and the estimated degrees of freedom reported by the command provides a useful diagnostic of how informative the collection is. Researchers should favour rich collections of outcomes measured for all units. Third, TMO composes naturally with existing practice: it can augment clustered, distance-based, and SCPC inference, adding precisely those pairs of locations that the baseline procedure's geography would overlook. In the county-level application considered here, these combinations behave as theory suggests: distance-based baselines capture local dependence, SCPC captures smooth global dependence, and TMO contributes strong, specific dependencies among—possibly distant—pairs.

Finally, we note two caveats. TMO is designed for settings where dependence is *sparse*: most pairs of locations are (approximately) uncorrelated, with meaningful dependence concentrated in relatively few pairs. In data exhibiting very strong, pervasive spatial dependence—of the form studied by Müller and Watson's recent work on spatial unit roots—sparsity fails by construction, and researchers should first diagnose and remove such trends before applying TMO or, indeed, most other spatial corrections. And as with all procedures discussed here, the number of auxiliary outcomes and locations drives computational cost; the command is written to remain practical at the scale of U.S. counties on a standard laptop.

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6 References

Acemoglu, D., S. Naidu, P. Restrepo, and J. A. Robinson. 2019. Democracy Does Cause Growth. *Journal of Political Economy* 127(1): 47–100.

Becker, S. O., P. D. Boll, and H.-J. Voth. 2026. Testing and Correcting for Spatial Unit Roots in Regression Analysis. *Stata Journal* 26(2): 177–202.

Cameron, A. C., and D. L. Miller. 2015. A Practitioner's Guide to Cluster-Robust Inference. *Journal of Human Resources* 50(2): 317–372.

Conley, T. 1999. GMM estimation with cross sectional dependence. *Journal of Econometrics* 92(1): 1–45.

DellaVigna, S., G. Imbens, W. Kim, and D. M. Ritzwoller. 2025. Using Multiple Outcomes to Adjust Standard Errors for Spatial Correlation. https://arxiv.org/abs/2504.13295.

Eicker, F. 1967. Limit theorems for regressions with unequal and dependent errors. In *Proceedings of the Fifth Berkeley Symposium on Mathematical Statistics and Probability*, vol. 1, 59–82. University of California Press.

Huber, P. J. 1967. The behavior of maximum likelihood estimates under nonstandard conditions. In *Proceedings of the Fifth Berkeley Symposium on Mathematical Statistics and Probability*, vol. 1, 221–233. University of California Press.

Liang, K.-Y., and S. L. Zeger. 1986. Longitudinal data analysis using generalized linear models. *Biometrika* 73(1): 13–22.

Long, J. S., and L. H. Ervin. 2000. Using Heteroscedasticity Consistent Standard Errors in the Linear Regression Model. *The American Statistician* 54(3): 217–224.

MacKinnon, J. G., and H. White. 1985. Some heteroskedasticity-consistent covariance matrix estimators with improved finite sample properties. *Journal of Econometrics* 29(3): 305–325.

Moulton, B. R. 1986. Random group effects and the precision of regression estimates. *Journal of Econometrics* 32: 385–397.

———. 1990. An Illustration of a Pitfall in Estimating the Effects of Aggregate Variables on Micro Units. *The Review of Economics and Statistics* 72(2): 334–338.

Müller, U. K., and M. W. Watson. 2022a. Spatial Correlation Robust Inference in Linear Regression and Panel Models. *Journal of Business & Economic Statistics* 41(4): 1050–1064.

———. 2022b. Spatial Correlation Robust Inference. *Econometrica* 90(6): 2901–2935.

Newey, W. K., and K. D. West. 1987. A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix. *Econometrica* 55: 703–708.

Rogers, W. H. 1993. sg17: Regression standard errors in clustered samples. *Stata Technical Bulletin* 13: 19–23. Reprinted in Stata Technical Bulletin Reprints, Vol. 3, pp. 88–94.

White, H. 1980. A heteroskedasticity-consistent covariance matrix estimator and a direct test for heteroskedasticity. *Econometrica* 48: 817–838.

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Appendix

Algorithm 1 Thresholding Multiple Outcomes (TMO)

Require: Outcome $Y_i^{(0)}$ and treatment W_i for units $i = 1, \ldots, n$

Ensure: Estimate of the variance of $\hat{\tau}^{(0)}$, the least-squares regression coefficient of $Y_i^{(0)}$ on W_i

1: Identify a collection of auxiliary outcomes $Y_i^{(1)}, \ldots, Y_i^{(d)}$.

2: For each unit i and outcome j, compute the normalized residuals $\hat{\varepsilon}_i^{(j)}$

3: For each pair of units $i \neq i'$, measure the correlation $\hat{\rho}_{i,i'}$

4: Approximate the null distribution of $\hat{\rho}_{i,i'}$ with $\mathrm{N}(0, \hat{v})$, where $\hat{v}$ solves

$$\Phi_{\hat{v}}^{-1}(0.75) - \Phi_{\hat{v}}^{-1}(0.25) \;=\; G_{n,d}^{-1}(0.75) - G_{n,d}^{-1}(0.25),$$

where $\Phi_v(\cdot)$ is the CDF of $\mathrm{N}(0, v)$, and $G_{n,d}(\cdot)$ is defined (paper)

5: Choose the threshold $\hat{\delta}^*$ by maximizing

$$\hat{Q}_{n,d}(\delta) \;=\; \big(\hat{F}_{n,d}(\delta) - F_0^+(\delta)\big) - F_0^+(\delta), \qquad \delta > 0,$$

where $F_0^+(\delta) = 2\big(1 - \Phi_{\hat{v}}(\cdot)\big)$ and $\hat{F}_{n,d}(\cdot)$ defined in (paper).

6: Collect the estimates

$$\hat{\sigma}_{i,i'}^{(0)}(\hat{\delta}^*) = \begin{cases} \hat{\varepsilon}_i^{(0)}\, \hat{\varepsilon}_{i'}^{(0)}, & i = i', \\ \hat{\varepsilon}_i^{(0)}\, \hat{\varepsilon}_{i'}^{(0)}\, \mathbb{I}\{\, |\hat{\rho}_{i,i'}| \geq \hat{\delta}^* \,\}, & i \neq i', \end{cases}$$

into the matrix $\hat{\Sigma}^{(0)}(\hat{\delta}^*)$.

7: **Return** the variance estimate

$$\hat{V}(\hat{\delta}^*) = \big(W^\top W\big)^{-1} \Big(W^\top\, \hat{\Sigma}^{(0)}(\hat{\delta}^*)\, W\Big) \big(W^\top W\big)^{-1}.$$